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PMC9086392 | Mizoroki – Heck 反应 | 采集状态 imported_unchecked

标题	Preparation of a novel palladium catalytic hydrogel based on graphene oxide/chitosan NPs and cellulose nanowhiskers †
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摘要（原文节选）

The present paper focuses on the synthesis of a novel hydrogel support by combining polysaccharides (chitosan NPs and dialdehyde cellulose nanowhiskers) and graphene oxide nanosheets to obtain a biocompatible material for catalytic applications. The hydrogel was synthesized via green chemistry processes and used as a support to prepare Pd nanoparticles. Finally, the hydrogel@Pd NPs was employed as the catalyst in the Mizoroki – Heck reaction to generate new C – C bonds. SEM analysis indicated that the hydrogel has macr ⋯

数据表 · 表 1（14 行，含数值行 14 行）

Optimization of reaction temperature for the model coupling reactiona

Entry	Pd (mol%)	Solvent	Base	Time (min.)	Yield (%)b
1	0.20	Toluene	K2CO3	60	75
2	0.20	DMF	K2CO3	50	79
3	0.20	EtOH	K2CO3	40	80
4	0.20	H2O	K2CO3	60	60
5	0.20	EtOH/H2O	K2CO3	35	96
6	0.20	EtOH/H2O	NaOAc	60	77
7	0.20	EtOH/H2O	NEt3	60	65
8	0.20	EtOH/H2O	NaHCO3	65	85
9	0.20	EtOH/H2O	t-BuOK	50	79
10	0.20	EtOH/H2O	No base	100	Trace
11	0.15	EtOH/H2O	K2CO3	35	83
12	0.10	EtOH/H2O	K2CO	40	75
13	0.25	EtOH/H2O	K2CO	30	90
14	0.0	EtOH/H2O	K2CO	120	0.0

脚注（逐字）：aReaction conditions: iodobenzene, methyl acrylate, catalyst, base and solvent.bIsolated yield.

正文对应句（逐字）：Initially, to evaluate the effect of the catalyst, the model reaction was performed in the absence and presence of different amounts of hydrogel nanocatalyst under reflux conditions. As expected, no product was obtained in the absence of the catalyst (entry 14); therefore, the addition of a catalyst

数据表 · 表 2（12 行，含数值行 12 行）

Mizoroki – Heck reaction coupling reaction of different aryl halides with olefinsa

Entry	Olefin	X	R1	Time (min)	Yield (%)b
1	Styrene	I	H	30	98
2	Methyl acrylate	I	H	35	96
3	Methyl acrylate	I	NH2	40	92
4	Methyl methacrylate	I	H	35	95
5	Acrylonitrile	I	H	30	99
6	Methyl methacrylate	Br	H	55	95
7	Styrene	Br	H	40	97
8	Acrylonitrile	Br	H	45	95
9	Acrylonitrile	Br	NH2	65	90
10	Methyl acrylate	Br	H	60	93
11	Styrene	Cl	H	120	87
12	Methyl methacrylate	Cl	H	180	81

脚注（逐字）：aReaction conditions: iodobenzene, methyl acrylate, catalyst, base and solvent.bIsolated yield.

正文对应句（逐字）：Initially, to evaluate the effect of the catalyst, the model reaction was performed in the absence and presence of different amounts of hydrogel nanocatalyst under reflux conditions. As expected, no product was obtained in the absence of the catalyst (entry 14); therefore, the addition of a catalyst

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